

Continued Fractions

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Definition

A simple continued fraction is an expression of the form

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \dots}}}$$

Where $a_0 \in \mathbb{Z}$ and $a_1, \dots, a_n \in \mathbb{N}$

We write a continued fraction compactly as $[a_0; a_1, a_2, a_3, \dots]$

Simple Example

Write $\frac{31}{12}$ as a continued fraction Step 1: Find the floor of $\frac{31}{12}$

$$\left\lfloor \frac{31}{12} \right\rfloor = 2$$

This is your a_0

Step 2: Subtract your original number by the floor calculated in step 1.

$$\frac{31}{12} - 2 = \frac{31}{12} - \frac{24}{12} = \frac{7}{12}$$

This will always give you a number between 0 and 1.

Step 3: Flip the fraction

$$\frac{7}{12} \rightarrow \frac{12}{7}$$

Step 4: Repeat steps 1-3 until you get a 1 as the numerator of the fraction in step 2.

Finishing the example:

$$\text{Floor : } \left\lfloor \frac{12}{7} \right\rfloor = 1 = a_1$$

$$\text{Subtract : } \frac{12}{7} - 1 = \frac{5}{7}$$

$$\text{Flip : } \frac{5}{7} \rightarrow \frac{7}{5}$$

Repeat

$$\text{Floor : } \left\lfloor \frac{7}{5} \right\rfloor = 1 = a_2$$

$$\text{Subtract : } \frac{7}{5} - 1 = \frac{2}{5}$$

$$\text{Flip : } \frac{2}{5} \rightarrow \frac{5}{2}$$

$$\text{Floor : } \left\lfloor \frac{5}{2} \right\rfloor = 2 = a_3$$

$$\text{Subtract : } \frac{5}{2} - 2 = \frac{1}{2}$$

Stop

We stop since 1 is in the numerator of the fraction.

Now using our a'_i s, we can write our continued fraction:

$$\frac{31}{12} = 2 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2 + \frac{1}{2}}}}$$

Irrational Example